

Module 4: Supervised Learning
Submodule: 4.7 Naive Baye
Lecture: Naive Bayes Spam Classification

XCS229 Machine Learning

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I. Introduction

In this section, we introduce the problem of **spam classification**, which involves determining whether a piece of text (an email) is spam or not. We outline the goal of developing a generative learning algorithm to tackle this classification problem. And we are going to have discrete x .

II. Feature Representation for Spam Classification

We need to represent the text in a numerical format, as computers process numerical data. The aim is to convert a piece of text into a feature vector x of dimension d , where d represents the size of the vocabulary.

$$\text{text} \longrightarrow x \in \mathbb{R}^d$$

In our approach to constructing a spam filter, we will begin by specifying the features x_j used to represent an email.

We will represent each email using a feature vector whose length is equal to the number of words in our dictionary. We list all words from our vocabulary (10,000 words) in alphabetical order. Specifically, if an email contains the j -th word from the dictionary, we will set $x_j = 1$; otherwise, we will set $x_j = 0$.

For instance, consider the following feature vector:

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \begin{array}{l} \text{a} \\ \text{aardvark} \\ \text{aardwolf} \\ \vdots \\ \text{buy} \\ \vdots \\ \text{zygmurgy} \end{array}$$

This vector x is used to represent an email containing the words “a” and “buy,” but not “aardvark,” “aardwolf” or “zygmurgy” (which is a branch of applied chemistry that deals with the science and practice of fermentation... but you already knew that). The set of words that is encoded into the feature vector is referred to as the **vocabulary**. Consequently, the dimension of the vector x is equal to the size of the vocabulary (i.e., $d = 10,000$).

So if we have a sentence like “I buy a book”, we create a vector $x \in \{0, 1\}^d$ where:

$$x_i = \begin{cases} 1 & \text{if } i^{\text{th}}\text{-word appears in the email} \\ 0 & \text{if } i^{\text{th}}\text{-word does not appear in the email} \end{cases}$$

We label each entry corresponding to a word in the vocabulary, leading to a representation that captures the presence or absence of words regardless of order or frequency. This is simplistic because we don’t capture the order of the words or the frequency with which they appear but this representation will simplify the math.

Now that we have established our feature vector x , which is a binary vector taking values of 0 or 1, we need a distribution that can generate these binary vectors. In this context, we will utilize the Naive Bayes classification technique.

III. Generative Model

In this section, we outline how to build a generative model for x given y . Here, y indicates whether the email is spam (1) or not spam (0).

Naive Bayes Assumption

The estimation of the joint distribution is done through the following factorization:

$$P(x|y) = P(x_1|y) \cdots P(x_d|y)$$

To model $p(x|y)$, we assume that $x_1, x_2, \dots, x_d$ are *conditionally independent* given y . This means that once we know y , we can independently draw each feature x_i .

“Conditional independence” is the **Naive Bayes (NB) assumption**, and the resulting algorithm is called a **Naive Bayes classifier**.

It is important to note that this independence assumption may not accurately reflect how people generate emails. As an example:

- When generating a spam email, one might first decide if the email is spam (determine y) and then randomly select which features to include in the email. This is likely not how actual spam emails are written.

However, despite the unrealistic nature of this assumption, it turns out that in many cases, such simplifying assumptions can lead to effective outcomes. For instance, utilizing this Naive Bayes model in practice can achieve over 90% accuracy in spam classification.

As of 2012, the effectiveness of this approach may have diminished due to the adversarial nature of spammers, who can modify their emails to evade detection algorithms. Nonetheless, this method was a reasonable approach around ten years ago.

Interestingly, even if the assumption of independence is not entirely correct, it may still yield useful results due to the partial correctness of the assumption. For our purposes, the focus is less on the validity of the assumptions and more on demonstrating the methodology of applying this probabilistic model effectively.

Another consideration is how the length of the text impacts representation. In our binary vector model, the length of the original text does not play a significant role. We can encode any length of text into a single vector of dimension d (the size of the vocabulary). This can be viewed as both an advantage and a disadvantage.

When deciding what constitutes the “window size” for our feature vector, it is generally practical to take the entire email as the unit of classification, as we are assessing whether the complete email is spam or not. Therefore, the classification will determine whether each emails qualifies as spam, rather than classifying individual sentences.

IV. Parameterization of the Naive Bayes Model

In the Naive Bayes classifier, we represent the joint probability of the features x given a class label y as follows:

$$P(x_1, \dots, x_d | y) = P(x_1 | y) \cdots P(x_d | y)$$

To understand this better, we need to parameterize each of these probability distributions using some parameters. Given that each feature x_i can only take on two values, 0 or 1, we note that these probabilities are typically modeled using a Bernoulli distribution. Therefore, for each feature, we can describe its distribution with a single parameter.

Let $\phi_{j|y=1} = P(x_j = 1 | y = 1)$ denote the probability of feature x_j being 1 when y is equal to 1 (indicating that the email is spam). Consequently, we have:

$$P(x_j = 0 | y = 1) = 1 - \phi_{j|y=1}$$

Similarly, we define another parameter:

$$\phi_{j|y=0} = P(x_j = 1 | y = 0)$$

which gives us:

$$P(x_j = 0 | y = 0) = 1 - \phi_{j|y=0}$$

Additionally, we need to parameterize the probability of the class label itself:

$$\phi_y = P(y = 1)$$

Here, ϕ_y serves to define the distribution of the class label y .

As we proceed, we will leverage these parameters to calculate the overall probabilities for classifying emails as spam using Naive Bayes.

V. Maximizing the Likelihood in Naive Bayes

In this section, we discuss how to write the likelihood function and subsequently maximize it.

The likelihood L is defined as the probability of observing the data given the parameters:

$$L(\phi_y, \phi_{1|y=0}, \dots, \phi_{d|y=0}, \phi_{1|y=1}, \dots, \phi_{d|y=1}) = \prod_{i=1}^n p(x^{(i)}, y^{(i)}; \phi_y, \phi_{j|y}).$$

Note, $\phi_y, \phi_{j|y}$ is just a convenient way to represent *all* the parameters, which are $\phi_y, \phi_{1|y=0}, \dots, \phi_{d|y=0}, \phi_{1|y=1}, \dots, \phi_{d|y=1}$.

The likelihood can be factored into the product of the probabilities of each example because all examples are assumed to be independent:

$$= \prod_{i=1}^n p(x^{(i)} | y^{(i)}; \phi_y, \phi_{j|y}) \cdot p(y^{(i)}; \phi_y, \phi_{j|y})$$

This can be further simplified using the chain rule:

$$= \prod_{i=1}^n \left(p(y^{(i)}; \phi_y) \cdot \prod_{j=1}^d p(x_j^{(i)} | y^{(i)}; \phi_{j|y}) \right)$$

The next step is to maximize this likelihood. We know that maximizing L is equivalent to maximizing $\log L$:

$$\arg \max L = \arg \max \log L$$

Converting the likelihood to its logarithmic form yields:

$$\log L(\phi_y, \phi_{j|y}) = \ell(\phi_y, \phi_{j|y}) = \sum_{i=1}^n \log p(y^{(i)}; \phi_y) + \sum_{i=1}^n \sum_{j=1}^d \log p(x_j^{(i)} | y^{(i)}; \phi_{j|y})$$

To find the maximum, we compute the gradient of the log-likelihood and set it to zero:

$$\nabla \ell(\phi_i, \phi_{j|y}) = 0.$$

Thus, solving the maximization yields parameter estimates as follows:

$$\begin{aligned} \phi_y &= \frac{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}}{n} \\ \phi_{j|y=1} &= \frac{\sum_{i=1}^n \mathbb{1}\{x_j^{(i)} = 1, y^{(i)} = 1\}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}} \\ \phi_{j|y=0} &= \frac{\sum_{i=1}^n \mathbb{1}\{x_j^{(i)} = 1, y^{(i)} = 0\}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\}} \end{aligned}$$

where $\mathbb{1}$ is the indicator function that represents whether the sample corresponds to a positive example. Here:

- ϕ_y represents the fraction of positive examples.
- $\phi_{j|y=1}$ gives the probability of the j^{th} word being 1 given the label y is 1 (spam), which can be interpreted as the count of occurrences of the j^{th} word across all positive examples, normalized by the total number of positive examples.
- Similarly, $\phi_{j|y=0}$ is defined for negative examples.

This shows that our model's estimates update based on the frequencies of the words in either class, making the method straightforward and intuitive.

VI. Summary

- **Spam Classification:** The goal is to determine whether an email is spam or not using a generative learning algorithm.
- **Feature Representation:** Text is converted into binary feature vectors, where each element represents the presence of a word from a predefined vocabulary. If the word is present, it is set to 1; otherwise, it is set to 0.
- **Generative Model:** The Naive Bayes model relies on the Naive Bayes assumption of conditional independence, meaning that each feature is independent given the class label (spam or not spam).
- **Model Parameterization:** The probabilities of features given the class are parameterized using terms like $\phi_{j|y=1}$ and $\phi_{j|y=0}$.
- **Maximizing the Likelihood:** The likelihood function L is maximized, and the logarithm of the likelihood is utilized for easier computation. Parameter estimates are derived from the frequency of words in the respective classes.
- **Practical Importance:** The Naive Bayes classifier can achieve good accuracy in spam detection, despite its simplifying assumptions.